

## SAFETY DATA SHEET

Creation Date 23-Jan-2009

Revision Date 26-Dec-2021

Revision Number 7

### 1. Identification

**Product Name** Methyl sulfoxide

**Cat No. :** AC428680000; AC428680025

**CAS No** 67-68-5  
**Synonyms** Dimethyl sulfoxide; DMSO

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

**Company**

Fisher Scientific Company  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

Acros Organics  
One Reagent Lane  
Fair Lawn, NJ 07410

**Emergency Telephone Number** For information **US** call: 001-800-ACROS-01 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

**Classification**

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

Flammable liquids

Category 4

**Label Elements**

**Signal Word**  
Warning

**Hazard Statements**  
Combustible liquid

**Precautionary Statements**  
Prevention

Keep away from heat/sparks/open flames/hot surfaces. - No smoking

**Fire**

In case of fire: Use CO<sub>2</sub>, dry chemical, or foam for extinction

**Storage**

Store in a well-ventilated place. Keep cool

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Hazards not otherwise classified (HNOC)**

None identified

**Other hazards**

DMSO readily penetrates skin and may carry other dissolved chemicals into the body.

### 3. Composition/Information on Ingredients

| Component          | CAS No  | Weight % |
|--------------------|---------|----------|
| Dimethyl sulfoxide | 67-68-5 | >95      |

### 4. First-aid measures

|                                            |                                                                                                                          |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                      | If symptoms persist, call a physician. Show this safety data sheet to the doctor in attendance.                          |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.          |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur.  |
| <b>Inhalation</b>                          | Remove to fresh air. Get medical attention immediately if symptoms occur. If not breathing, give artificial respiration. |
| <b>Ingestion</b>                           | Do NOT induce vomiting. Get medical attention.                                                                           |
| <b>Most important symptoms and effects</b> | Difficulty in breathing. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting             |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                                                                    |

### 5. Fire-fighting measures

|                                         |                                                                                                                                         |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Water spray, carbon dioxide (CO <sub>2</sub> ), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                                                                                |
| <b>Flash Point</b>                      | 87 °C / 188.6 °F                                                                                                                        |
| <b>Method -</b>                         | No information available                                                                                                                |
| <b>Autoignition Temperature</b>         | 301 °C / 573.8 °F                                                                                                                       |
| <b>Explosion Limits</b>                 |                                                                                                                                         |
| <b>Upper</b>                            | 42 vol %                                                                                                                                |
| <b>Lower</b>                            | 2.6 vol %                                                                                                                               |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                                                                |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                                                                |

**Specific Hazards Arising from the Chemical**

Combustible material. Containers may explode when heated. Thermal decomposition can lead to release of irritating gases and vapors.

**Hazardous Combustion Products**

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Sulfur oxides. Sulfides. Formaldehyde.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**NFPA**

**Health**  
2 \*

**Flammability**  
2

**Instability**  
1

**Physical hazards**  
N/A

**6. Accidental release measures****Personal Precautions**

Use personal protective equipment as required. Remove all sources of ignition. Take precautionary measures against static discharges. Ensure adequate ventilation.

**Environmental Precautions**

Should not be released into the environment. Do not flush into surface water or sanitary sewer system. See Section 12 for additional Ecological Information.

**Methods for Containment and Clean Up**

Remove all sources of ignition. Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

**7. Handling and storage****Handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Keep away from open flames, hot surfaces and sources of ignition. Avoid contact with skin, eyes or clothing. Avoid ingestion and inhalation.

**Storage.**

Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame. Incompatible Materials. Strong oxidizing agents. Strong acids. Strong bases. Alkali metals.

**8. Exposure controls / personal protection****Exposure Guidelines**

This product does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

**Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

**Personal Protective Equipment****Eye/face Protection**

Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

**Skin and body protection**

Wear appropriate protective gloves and clothing to prevent skin exposure.

**Respiratory Protection**

Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

**9. Physical and chemical properties****Physical State**

Liquid

|                                        |                                   |
|----------------------------------------|-----------------------------------|
| Appearance                             | Colorless                         |
| Odor                                   | Odorless                          |
| Odor Threshold                         | No information available          |
| pH                                     | No information available          |
| Melting Point/Range                    | 18.4 °C / 65.1 °F                 |
| Boiling Point/Range                    | 189 °C / 372.2 °F                 |
| Flash Point                            | 87 °C / 188.6 °F                  |
| Evaporation Rate                       | No information available          |
| Flammability (solid,gas)               | Not applicable                    |
| Flammability or explosive limits       |                                   |
| Upper                                  | 42 vol %                          |
| Lower                                  | 2.6 vol %                         |
| Vapor Pressure                         | 0.55 mbar @ 20°C                  |
| Vapor Density                          | 2.7                               |
| Specific Gravity                       | 1.100                             |
| Solubility                             | Soluble in water                  |
| Partition coefficient; n-octanol/water | No data available                 |
| Autoignition Temperature               | 301 °C / 573.8 °F                 |
| Decomposition Temperature              | > 190°C                           |
| Viscosity                              | 1.98 mPa.s @ 25°C                 |
| Molecular Formula                      | C <sub>2</sub> H <sub>6</sub> O S |
| Molecular Weight                       | 78.13                             |

## 10. Stability and reactivity

|                                  |                                                                                                                                       |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                                                                                            |
| Stability                        | Hygroscopic.                                                                                                                          |
| Conditions to Avoid              | Incompatible products. Excess heat. Exposure to moist air or water. Keep away from open flames, hot surfaces and sources of ignition. |
| Incompatible Materials           | Strong oxidizing agents, Strong acids, Strong bases, Alkali metals                                                                    |
| Hazardous Decomposition Products | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), Sulfur oxides, Sulfides, Formaldehyde                                        |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                                                              |
| Hazardous Reactions              | Thermal decomposition can take place above 189°C / 372°F.                                                                             |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

#### Component Information

| Component          | LD50 Oral                  | LD50 Dermal                | LC50 Inhalation              |
|--------------------|----------------------------|----------------------------|------------------------------|
| Dimethyl sulfoxide | LD50 = 28300 mg/kg ( Rat ) | LD50 = 40000 mg/kg ( Rat ) | LC50 > 5.33 mg/L ( Rat ) 4 h |

**Toxicologically Synergistic Products** No information available

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Irritation      | No information available                                                                 |
| Sensitization   | No information available                                                                 |
| Carcinogenicity | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component          | CAS No  | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|--------------------|---------|------------|------------|------------|------------|------------|
| Dimethyl sulfoxide | 67-68-5 | Not listed | Not listed | Not listed | Not listed | Not listed |

|                                                   |                                                                                     |
|---------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>Mutagenic Effects</b>                          | No information available                                                            |
| <b>Reproductive Effects</b>                       | No information available.                                                           |
| <b>Developmental Effects</b>                      | No information available.                                                           |
| <b>Teratogenicity</b>                             | No information available.                                                           |
| <b>STOT - single exposure</b>                     | None known                                                                          |
| <b>STOT - repeated exposure</b>                   | None known                                                                          |
| <b>Aspiration hazard</b>                          | No information available                                                            |
| <b>Symptoms / effects, both acute and delayed</b> | Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting |
| <b>Endocrine Disruptor Information</b>            | No information available                                                            |
| <b>Other Adverse Effects</b>                      | The toxicological properties have not been fully investigated.                      |

## 12. Ecological information

### Ecotoxicity

Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants. Do not empty into drains. .

| Component          | Freshwater Algae            | Freshwater Fish                         | Microtox                                                                                                                                              | Water Flea         |
|--------------------|-----------------------------|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| Dimethyl sulfoxide | EC50 96h 12350 - 25500 mg/L | 40 g/L LC50 96 h<br>33-37 g/L LC50 96 h | = 16000 mg/L EC50<br>Pseudomonas putida 16 h<br>= 32 g/L EC50 Tetrahymena<br>pyriformis 24 h<br>= 77 mg/L EC50<br>Photobacterium<br>phosphoreum 5 min | EC50 24h 7000 mg/L |

|                                      |                                                                         |
|--------------------------------------|-------------------------------------------------------------------------|
| <b>Persistence and Degradability</b> | Persistence is unlikely                                                 |
| <b>Bioaccumulation/ Accumulation</b> | No information available.                                               |
| <b>Mobility</b>                      | . Will likely be mobile in the environment due to its water solubility. |

| Component          | log Pow |
|--------------------|---------|
| Dimethyl sulfoxide | -2.03   |

## 13. Disposal considerations

|                               |                                                                                                                                                                                                                                                                 |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste Disposal Methods</b> | Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification. |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 14. Transport information

|                             |                                                                                                                                                                                                           |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DOT</b>                  | COMBUSTIBLE LIQUID, NOT REGULATED FOR TRANSPORT IN THIS QUANTITY<br>According to 49 CFR §173.150(f)(1), this material should be reclassified as NA1993, Combustible Liquid, NOS if it is shipped in bulk. |
| <b>UN-No</b>                | NA1993                                                                                                                                                                                                    |
| <b>Proper Shipping Name</b> | Combustible liquid, n.o.s.                                                                                                                                                                                |
| <b>Packing Group</b>        | III                                                                                                                                                                                                       |
| <b>TDG</b>                  | Not regulated                                                                                                                                                                                             |
| <b>IATA</b>                 | Not regulated                                                                                                                                                                                             |
| <b>IMDG/IMO</b>             | Not regulated                                                                                                                                                                                             |

## 15. Regulatory information

**United States of America Inventory**

| Component          | CAS No  | TSCA | TSCA Inventory notification - Active-Inactive | TSCA - EPA Regulatory Flags |
|--------------------|---------|------|-----------------------------------------------|-----------------------------|
| Dimethyl sulfoxide | 67-68-5 | X    | ACTIVE                                        | -                           |

**Legend:**

TSCA US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

- - Not Listed

TSCA 12(b) - Notices of Export Not applicable

**International Inventories**

Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component          | CAS No  | DSL | NDSL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL     |
|--------------------|---------|-----|------|-----------|-------|------|------|------|-------|----------|
| Dimethyl sulfoxide | 67-68-5 | X   | -    | 200-664-3 | X     | X    | X    | X    | X     | KE-32367 |

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)**U.S. Federal Regulations**

SARA 313 Not applicable

SARA 311/312 Hazard Categories See section 2 for more information

CWA (Clean Water Act) Not applicable

Clean Air Act Not applicable

OSHA - Occupational Safety and Health Administration Not applicable

CERCLA Not applicable

California Proposition 65 This product does not contain any Proposition 65 chemicals.

**U.S. State Right-to-Know Regulations**

| Component          | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|--------------------|---------------|------------|--------------|----------|--------------|
| Dimethyl sulfoxide | -             | X          | -            | -        | -            |

**U.S. Department of Transportation**

Reportable Quantity (RQ): N

DOT Marine Pollutant N

DOT Severe Marine Pollutant N

U.S. Department of Homeland Security This product does not contain any DHS chemicals.

**Other International Regulations**

Mexico - Grade Slight risk, Grade 1

Authorisation/Restrictions according to EU REACH

| Component          | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|--------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Dimethyl sulfoxide | -                                                                   | Use restricted. See item 75.<br>(see link for restriction details)            | -                                                                                                     |

<https://echa.europa.eu/substances-restricted-under-reach>

#### Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component          | CAS No  | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|--------------------|---------|----------|------------------------------|---------------------------|--------------------------------------------|
| Dimethyl sulfoxide | 67-68-5 | Listed   | Not applicable               | Not applicable            | Not applicable                             |

| Component          | CAS No  | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|--------------------|---------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Dimethyl sulfoxide | 67-68-5 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

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**Revision Summary** This document has been updated to comply with the US OSHA HazCom 2012 Standard replacing the current legislation under 29 CFR 1910.1200 to align with the Globally Harmonized System of Classification and Labeling of Chemicals (GHS).

#### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**